

AgroScan Sentinel Platform

Full-Stack Technical Architecture, Framework Specifications, Database Design, Neural Diagnostics & System Build Manual

SYSTEM VERSION

v2.4.0 (Enterprise)

FRONTEND STACK

React 18 + Vite 5 + Tailwind

BACKEND STACK

Node.js + Express REST

DATABASE MODEL

Hybrid Document DB + Mongo

1. Executive Summary & Core Capabilities

AgroScan Sentinel is an advanced AI-powered Agricultural Decision Support System (DSS) engineered to deliver real-time crop disease diagnosis, multispectral leaf pathology classification, IoT sensor microclimate telemetry, GPS geospatial outbreak mapping, and agronomist consultation dispatching.



Neural Leaf Diagnostic Scanner

Simulated deep learning foliar analysis classifying phytopathogens against 45,000+ botanical models with localized lesion bounding boxes and severity indexing.



Live GPS Geospatial Sentinel

HTML5 browser geolocation + dual reverse geocoding resolving exact farm area names with layered cartography (Standard, Satellite Earth, NDVI vegetation canopy).



Dual-Method Authentication Gateway

1-Click Google Workspace (Gmail) SSO and Mobile Phone Number + SMS 4-Digit OTP verification with strict scanner access gating.



Animated Daylight / Dark Mode

Smooth 180° animated spin theme toggle between high-contrast Daylight Mode (☀️) and deep obsidian Dark Sentinel Mode (🌑).

2. Comprehensive Technology Stack & Frameworks

LAYER / SUBSYSTEM	TECHNOLOGY	VERSION	DESCRIPTION & TECHNICAL ROLE
Frontend Core	React	18.3.1	Single Page Application (SPA) reactive UI components, hooks (useState, useEffect, useMemo, useRef).
Bundler & Dev Server	Vite	5.4.21	High-speed Hot Module Replacement (HMR), ES module compilation, and Rollup production bundling.
Styling & CSS	Tailwind CSS	3.4.17	Utility-first CSS engine with custom botanical color palettes (forest, agri), glassmorphism, and dark mode class strategy.
CSS Processing	PostCSS / Autoprefixer	8.4.49	Automated CSS transformation pipeline with vendor prefixing.
Iconography	Lucide React	0.468.0	Featherweight, tree-shakeable SVG icon components.
Backend Runtime	Node.js	18+	High-performance asynchronous JavaScript server runtime using native ES module syntax (import/export).
Server Framework	Express.js	4.21.2	REST API microservice architecture, CORS negotiation, body parsing, and streaming file downloads.
Database (Local)	JSON Document DB	Native	Atomic, file-persisted JSON document store (server_db.json) requiring zero external setup.
Database (Cloud)	MongoDB + Mongoose	8.9.0	Object Data Modeling (ODM) layer providing schema validation and cluster connectivity.
Cartography & GIS	OpenStreetMap	Tile Layer API	Interactive satellite, street, and NDVI vegetation index map layers.
Reverse Geocoding	Nominatim & BigDataCloud	Open APIs	Resolves latitude/longitude coordinates into real-time Area, City, State, and Country names.

3. System Architecture & Data Flow

The platform follows a modular client-server decoupled architecture designed for high availability, offline capability, and low latency:

```
[ CLIENT SPA (React 18 + Vite) ]
├─ UI Views: Dashboard, Analysis Report, Field Map, Crop Encyclopedia, Community Hub
├─ Security & Auth: Gmail SSO & Mobile SMS OTP Gate
├─ Geolocation Engine: HTML5 Navigator GPS + Nominatim Geocoder
├─ State Manager: agroscoreStore.js (LocalStorage Reactive Store)
  ↓ (HTTP REST / JSON over Port 5000)
[ BACKEND SERVER (Node.js + Express) ]
├─ Auth Controller: /api/auth/login, /api/auth/register
├─ Neural AI Scan Engine: /api/scan (Multispectral classification simulation)
├─ Telemetry & Sensor Pipeline: /api/sensors, /api/sensors/update
├─ Outbreak Notification Dispatcher: /api/alerts
├─ Distribution Service: /api/download-zip (Live ZIP distribution)
  ↓
[ HYBRID DATABASE ROUTER ]
├─ Primary File Store: server_db.json (Zero-Setup Portability)
├─ Cluster Store: MongoDB Atlas / Local via Mongoose ODM
```

4. Core Functional Modules Deep-Dive

4.1. AI Leaf Diagnostic Scanner (QuickScanModal.jsx & AnalysisReportScreen.jsx)

- **Specimen Ingestion:** Supports high-definition photo upload, live camera feed, or calibrated field specimens.
- **Neural Inference Simulation:** Computes foliar chlorophyll spectral bands, lesion density, and pattern matches against 45,000+ pathogen models.
- **Lesion Bounding Boxes:** Overlays color-coded bounding boxes highlighting primary necrotic lesions and spore clusters with confidence percentages (e.g. 98.4%).
- **Prescriptive Action Guides:** Generates multi-step systemic chemical fungicide sprays, organic soil amendments, and perimeter quarantine instructions.

4.2. Live GPS Geospatial Field Sentinel (FieldMapScreen.jsx)

- **Browser GPS Geolocation:** Queries device latitude/longitude with high precision.
- **Real-Time Area Resolution:** Queries dual reverse geocoding services to display the exact farm name and locality (e.g. "Sector 4B • Iowa Agro-Corridor").
- **Layered Radar Map:** Allows switching between Standard Street, High-Res Satellite Earth, and NDVI vegetation false-color layers.

4.3. Dual-Method Authentication Gateway (AuthModal.jsx)

- **Gmail 1-Click Login:** Instant Google Workspace authentication.
- **Mobile Number + SMS OTP:** Country code selector (IN +91, US +1, GB +44, etc.), phone number entry, and 4-digit verification code with countdown timer.
- **Scanner Access Gating:** The diagnostic scanner is locked until authenticated; logging in automatically unlocks and launches the scanner.

5. REST API Specification Table

METHOD	ENDPOINT	PURPOSE	SAMPLE PAYLOAD / RESPONSE
GET	/api/health	Health check & active DB status	{ status: "online", database: {...} }
POST	/api/auth/login	Authenticates User ID / Gmail	{ emailOrId, password } → { success, user }
POST	/api/auth/register	Registers new farm account	{ email, password, name, role }
GET	/api/sensors	Retrieves sensor telemetry	{ fieldHealth, temperature, soilMoisture, ... }
POST	/api/sensors/update	Updates live sensor telemetry	{ ...sensorPayload } → { success, sensors }
GET	/api/reports	Fetches diagnostic scan reports	[{ id, code, diseaseName, confidence, ... }]
POST	/api/scan	Runs neural scan & creates report	{ sampleName, crop, previewUrl } → { report }
GET	/api/alerts	Returns active outbreak alerts	[{ id, title, type, description }]
POST	/api/expert-review	Submits agronomist review ticket	{ reportId, notes } → { ticketId }
GET	/api/download-zip	Streams full project ZIP archive	Binary stream (application/zip)
GET	/api/download-pdf	Streams the PDF technical manual	Binary stream (application/pdf)

6. Complete Project Directory Structure

```
crop-disease-platform/
├── src/
│   ├── components/
│   │   ├── AIWorkplaceWidget.jsx      # Floating AI assistant corner widget
│   │   ├── AnalysisReportScreen.jsx   # Detailed diagnostic report view
│   │   ├── AuthModal.jsx              # Official Gmail & Mobile OTP login portal
│   │   ├── BottomNav.jsx              # Sticky mobile bottom dock navigation
│   │   ├── CommunityScreen.jsx        # Agronomist Q&A discussion forum
│   │   ├── CropDiseaseView.jsx        # Disease encyclopedia with HD photos
│   │   ├── DashboardScreen.jsx        # Primary dashboard with in-page login & telemetry
│   │   ├── ExpertReviewModal.jsx      # Consultation ticket generator
│   │   ├── FertilizersView.jsx        # N-P-K nutrient & fertilizer guide
│   │   ├── FieldMapScreen.jsx         # Live GPS map with reverse geocoding
│   │   ├── Header.jsx                 # Top navigation bar with animated theme toggle
│   │   ├── QuickScanModal.jsx         # AI leaf camera & image analysis modal
│   │   ├── RegionalCropsView.jsx      # Regional crop recommendation catalog
│   │   ├── SettingsModal.jsx          # Farm settings & data reset modal
│   │   ├── Sidebar.jsx                # Desktop fixed navigation & drone HUD
│   │   └── WeatherModal.jsx           # 7-day microclimate forecast modal
│   ├── data/
│   │   ├── agroscanData.js            # Core sensors, reports, and leaf samples
│   │   └── cropData.js                # Disease taxonomy, fertilizers & regional data
│   ├── utils/
│   │   └── agroscanStore.js           # Reactive LocalStorage manager
│   ├── App.jsx                        # Root component & theme controller
│   └── main.jsx                       # React DOM entrypoint
├── server.js                          # Node.js + Express REST API backend
├── server_db.json                     # Persistent local JSON document database
├── tailwind.config.js                 # Tailwind CSS theme & dark mode config
├── vite.config.js                     # Vite bundler configuration
└── PROJECT_DOCUMENTATION.md          # Full project markdown documentation
```

7. Installation, Setup & Execution

To run the AgroScan Sentinel full-stack platform on any operating system:

```
# 1. Install all dependencies
npm install

# 2. Start the Backend API Server (Port 5000)
node server.js

# 3. Start the Frontend Development Server (Port 3000)
npm run dev -- --host --port 3000

# 4. Open in your browser
http://localhost:3000/
```